SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

COURSE: MASTER OF SCIENCE IN INFORMATION TECHNOLOGY

MIT 8103: DATA SCIENCE CONCEPTS

TOPIC: DIABETES PREDICTION

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PART 1: Problem Framing

1.1 Problem Definition

This analysis constructs a supervised binary classification model to predict whether a female patient is likely to have diabetes, given eight routine clinical and demographic measurements. The dataset contains 768 observations and a binary Outcome variable (1 = diabetic, 0 = non-diabetic). The model outputs a posterior probability per patient; a tuned threshold then converts this to a class label.

A false negative (diabetic patient classified as healthy) carries substantially greater clinical harm than a false positive (an unnecessary follow-up test). This asymmetric cost governs every methodological decision: metric selection, model choice, and threshold tuning.

1.2 Stakeholders

- Patients: primary beneficiaries of early detection; also bear the burden of false positives through unnecessary anxiety and procedures.
- Clinical providers (GPs, nurses, community health workers): use model outputs as a triage instrument to flag high-risk individuals, not as a diagnostic substitute.
- Public health authorities and policymakers: require scalable tools to enable risk-stratified resource allocation across large populations.
- Health economists and hospital administrators: concerned with reducing downstream costs from unmanaged diabetes (specialist care, hospitalization, surgical intervention).

1.3 Business and Policy Relevance

Dialysis, laser photocoagulation, and lower-limb amputation each cost orders of magnitude more than early intervention. A screening model requiring only routine clinical measurements already collected at check-ups can integrate into existing workflows at minimal cost, making it especially valuable where specialist referral capacity is limited.

1.4 Context

The dataset originates from a study of Pima Native American women aged 21 and above, a population with one of the highest documented Type 2 diabetes prevalence rates globally. This demographic specificity is a critical constraint: the model is validated solely on this population and must not be deployed to other groups without retraining.

PART 2: Data Understanding and Wrangling

2.1 Variable Descriptions and Data Types

Variable	R Type	Clinical Description
Pregnancies	Integer	Number of times pregnant
Glucose	Integer	2-hour plasma glucose (mg/dL)
BloodPressure	Integer	Diastolic blood pressure (mm Hg)
SkinThickness	Integer	Triceps skinfold thickness (mm)
Insulin	Integer	2-hour serum insulin (mu U/ml)
BMI	Numeric	Body mass index (kg/m2)

Variable	R Type	Clinical Description
DiabetesPedigreeFunction	Numeric	Genetic diabetes risk score
Age	Integer	Age in years
Outcome	Integer (Binary)	Target: 1=diabetic, 0=non-diabetic

2.2 Missing Values

Five columns use zero as a biological impossibility placeholder. These zeros were replaced with NA and imputed using the column median (preferred over mean because Insulin is heavily right-skewed).

```
zero_cols <- c('Glucose','BloodPressure','SkinThickness','Insulin','BMI')
df[zero_cols] <- lapply(df[zero_cols], function(x) ifelse(x==0, NA, x))
for (col in zero_cols) df[[col]][is.na(df[[col]])] <- median(df[[col]], na.rm=TRUE)
```

Column	Zero Count	Median Imputed With
Glucose	5	117 mg/dL
BloodPressure	35	72 mm Hg
SkinThickness	227	29 mm
Insulin	374	125 mu U/ml
BMI	11	32.3 kg/m2

2.3 Outlier Handling

IQR-based Winsorizing (manual implementation, 5th-95th percentile) caps extreme values in Insulin, BMI, DiabetesPedigreeFunction, and BloodPressure without removing any of the 768 rows.

```
winsorize_manual <- function(x, lo=0.05, hi=0.95) {
  pmax(pmin(x, quantile(x,hi)), quantile(x,lo)) }
for (col in c('Insulin','BMI','DiabetesPedigreeFunction','BloodPressure'))
  df[[col]] <- winsorize_manual(df[[col]])
```

2.4 Inconsistencies

The Outcome variable has a 65.1% / 34.9% class imbalance (500 non-diabetic, 268 diabetic). A classifier that always predicts the majority class would achieve 65.1% accuracy while being clinically worthless. This drives evaluation toward sensitivity-focused metrics. Outcome is converted to a factor for classification in R.

2.5 Feature Engineering

Three new features were created from clinical domain logic:

```
df$BMI_Category     <- cut(df$BMI, c(0,18.5,25,30,Inf), labels=0:3)  # WHO thresholds
df$Glucose_BMI_Score <- (df$Glucose * df$BMI) / 100                  # Compound risk
df$Young_High_Preg   <- as.integer(df$Age < 30 & df$Pregnancies > 3) # Gestational risk
```

BMI_Category captures WHO obesity thresholds (0=underweight, 1=normal, 2=overweight, 3=obese) that continuous BMI cannot express. Glucose_BMI_Score captures the multiplicative metabolic risk of simultaneous hyperglycaemia and obesity. Young_High_Preg flags elevated gestational diabetes risk in younger women with multiple pregnancies.

PART 3: Exploratory Data Analysis

3.1 Summary Statistics

Metric	Non-Diabetic (n=500)	Diabetic (n=268)	Difference
Mean Glucose (mg/dL)	110.68	142.13	+31.45
Mean BMI	30.93	34.94	+4.01
Mean Age (years)	31.19	37.07	+5.88
Mean Insulin (mu U/ml)	124.15	152.28	+28.13
Mean Blood Pressure (mm Hg)	70.85	74.60	+3.75
Mean Pregnancies	3.30	4.87	+1.57
Mean DPF Score	0.420	0.527	+0.107

3.2 Visualization 1: Feature Distributions (Histograms)

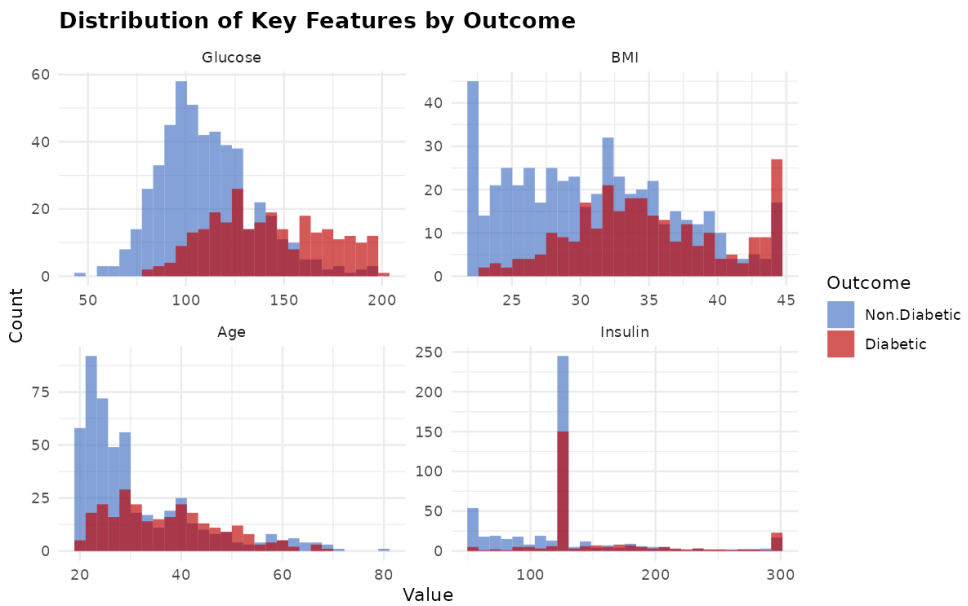

Figure 0-1 Glucose shows the strongest group separation; Insulin is right-skewed due to 48.7% imputation

3.3 Visualization 2: Feature Spread (Box Plots)

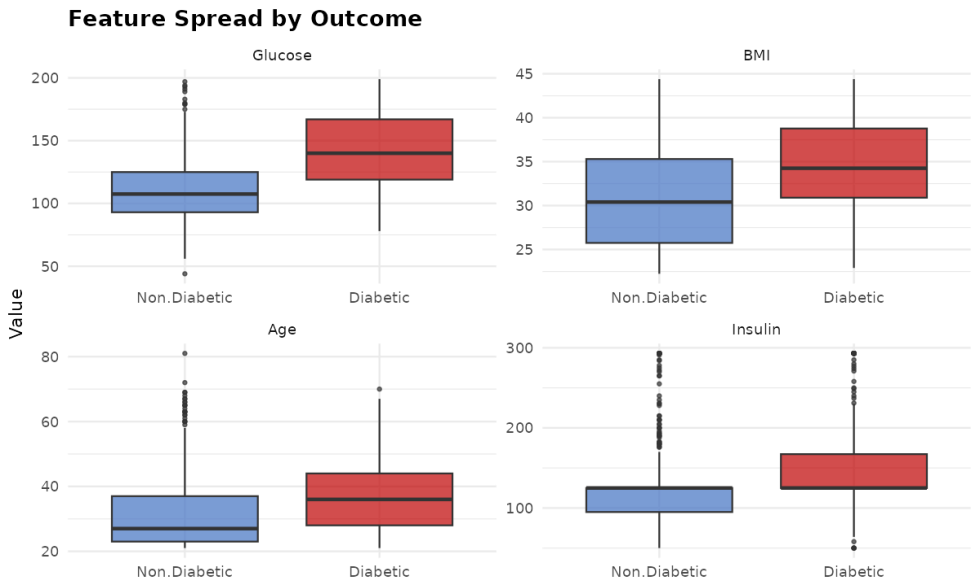

Figure 0-2 Diabetic glucose median sits well above the non-diabetic upper quartile.

3.4 Visualization 3: Class Distribution

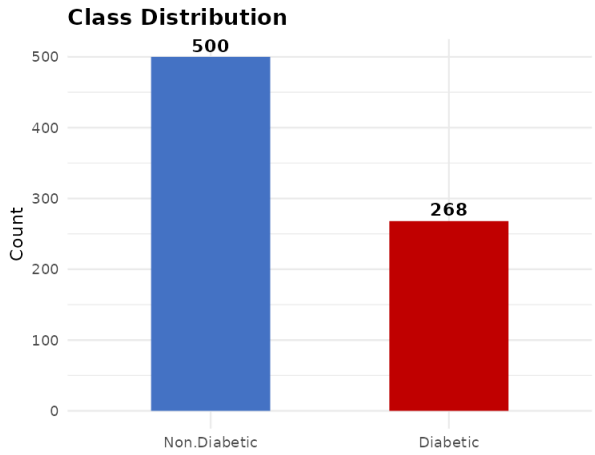

Figure 0-3 65.1% non-diabetic (500) vs 34.9% diabetic (268). Accuracy alone is misleading

3.5 Visualization 4: Glucose vs BMI Scatter Plot

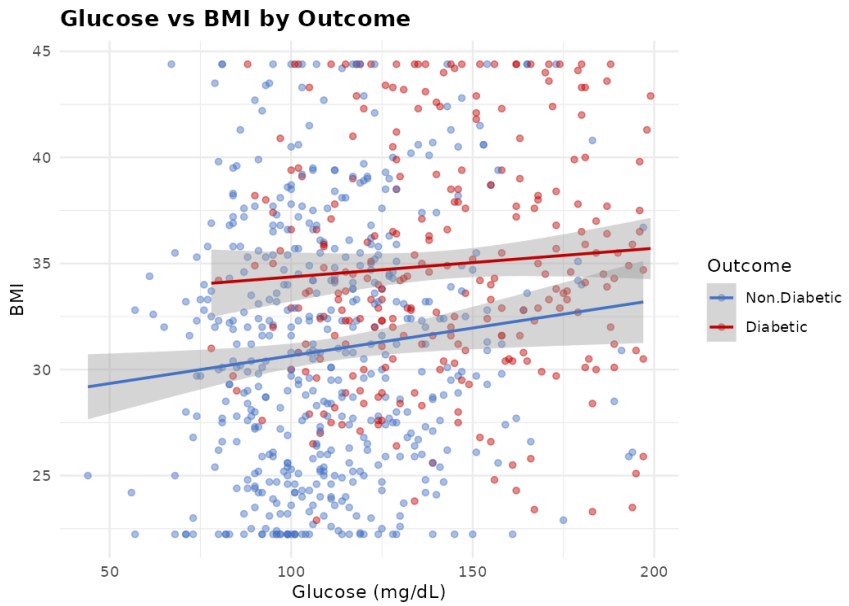

Figure 0-4 Diabetic patients cluster in the high-glucose, high-BMI quadrant, validating Glucose_BMI_Score

3.6 Visualization 5: Age Density Plot

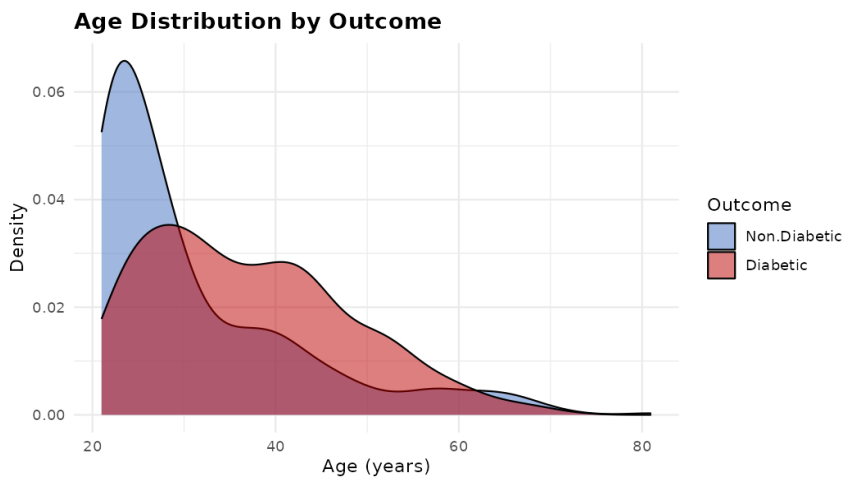

Figure 0-5 Non-diabetic patients peak sharply at ages 22-28; diabetic distribution extends into the 50s.

3.7 Correlation Analysis

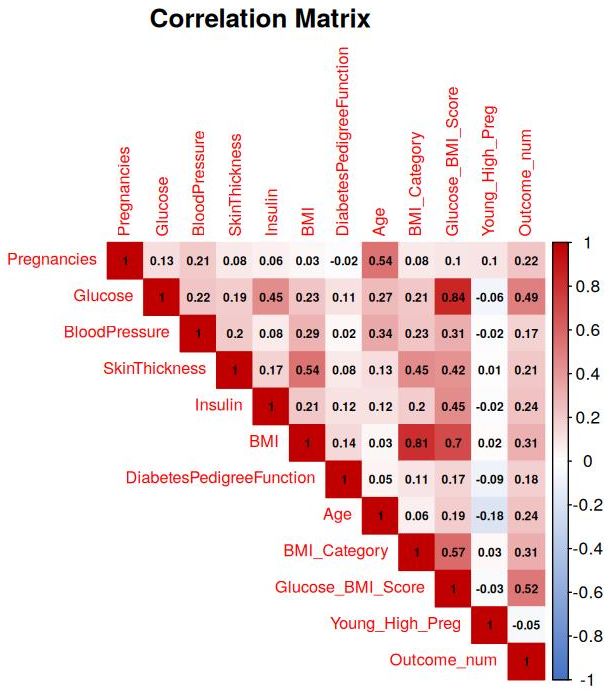

Figure 0-6 Correlation matrix. Glucose_BMI_Score (r=0.52) is the strongest predictor.

Feature	r with Outcome	Interpretation
Glucose_BMI_Score	0.5246	Strongest — engineered feature outperforms all raw variables
Glucose	0.4928	Close second; core clinical signal
BMI_Category	0.3105	Encoded WHO categories add ordinal information
BMI	0.3085	Strong secondary predictor
Insulin	0.2387	Attenuated by 48.7% imputation
Age	0.2384	Moderate; risk accelerates from 35
Pregnancies	0.2219	Moderate; gestational history
SkinThickness	0.2149	Moderate after imputation
DiabetesPedigreeFunction	0.1836	Genetic risk captured
BloodPressure	0.1713	Weak discriminator in this dataset

3.8 Key Insights

Insight 1: Glucose_BMI_Score is the Dominant Predictor

The engineered interaction feature (r=0.52) outperforms both its constituent variables, Glucose (r=0.49) and BMI (r=0.31), confirming that obesity and hyperglycaemia compound risk multiplicatively rather than additively. This validates the feature engineering decision and indicates the two conditions share underlying metabolic pathways.

Insight 2: Age Risk is Non-Linear

The density plot shows diabetes prevalence accelerating sharply from age 35, not gradually from 21. Uniform screening intervals are therefore inefficient; resources should concentrate in the 35-and-above cohort where the risk gradient is steepest.

Insight 3: Insulin Data Quality Limits Its Signal

Despite insulin resistance being central to Type 2 diabetes, Insulin's correlation (r=0.24) is modest. With 374 of 768 values (48.7%) replaced by the same median constant, the column carries no individual-patient information for nearly half the sample. This is a data quality limitation, not a biological one.

PART 4: Probability and Statistical Analysis

4.1 Random Variables

Each observation is a random draw from Pima Indian women aged 21+. The primary random variable, Outcome, follows a Bernoulli distribution with p = 268/768 = 0.349.

Variable	Type	Justification
Outcome	Discrete (Bernoulli)	Binary: 0 or 1 only
Pregnancies, Age	Discrete	Count / whole-year recording
Glucose, BMI, BP, Insulin	Continuous	Physiological measurements
DiabetesPedigreeFunction	Continuous	Computed decimal risk score

4.2 Marginal Probabilities

Event	Calculation	Probability
P(Diabetic)	268 / 768	0.3490
P(Non-Diabetic)	500 / 768	0.6510

4.3 Conditional Probability

Formally: P(A|B) = P(A and B) / P(B). Applied at the clinical 140 mg/dL glucose threshold:

P(Diabetic | Glucose >= 140) = count(Diabetic AND Glucose>=140) / count(Glucose>=140)

Condition	Computed Probability	Interpretation
P(Diabetic | Glucose >= 140 mg/dL)	0.6853	69% of high-glucose patients are diabetic
P(Diabetic | Glucose < 140 mg/dL)	0.2329	Only 23% of normal-glucose patients are diabetic
Baseline P(Diabetic)	0.3490	Without glucose information
Risk multiplier	2.94x	High glucose triples the posterior probability

Knowing a patient has glucose >= 140 mg/dL shifts the prior probability from 34.9% to 68.5%, nearly doubling diagnostic likelihood. This powerful Bayesian update directly motivates the 140 mg/dL screening trigger in Part 6.

4.4 Joint Probability

P(Diabetic AND Glucose >= 140 AND BMI >= 30) = 0.1523. This highest-risk subgroup (15.2% of the sample) should be the primary target for intensive clinical intervention.

4.5 Bayes' Theorem

P(Diabetic | Obese) = [P(Obese | Diabetic) x P(Diabetic)] / P(Obese)

Component	Computed Value
P(Obese | Diabetic)	0.8246
P(Obese) in full sample	0.6289
P(Diabetic) baseline	0.3490
P(Diabetic | Obese)	0.4576
Increase over baseline	+31.1%

Obesity alone raises the diabetes probability from 34.9% to 45.8%, a 31% increase, formalising the compound risk finding from the EDA.

4.6 Hypothesis Tests

Test 1: One-Tailed Welch t-Test on Glucose

H0: Mean glucose equal in both groups. H1: Higher in diabetic patients.

Statistic	Computed Value
t-statistic	14.8527
p-value	1.77 x 10^-41
Mean Glucose (Diabetic)	142.13 mg/dL
Mean Glucose (Non-Diab.)	110.68 mg/dL
Decision	Reject H0

Test 2: Two-Tailed Welch t-Test on BMI

H0: Mean BMI equal in both groups. H1: Mean BMI differs.

Statistic	Computed Value
t-statistic	9.2381
p-value	4.44 x 10^-19
Mean BMI (Diabetic)	34.94
Mean BMI (Non-Diab.)	30.93
Cohen's d	0.6892 (medium-large effect)
Decision	Reject H0

Cohen's d = 0.69 confirms the difference is not only statistically significant but clinically meaningful. Effect size is essential: a large sample can produce a statistically significant but practically trivial difference.

Test 3: Chi-Square on Glucose Category and Outcome

H0: Glucose category and Outcome are independent. H1: They are associated.

Statistic	Computed Value
Chi-square statistic	129.9382
Degrees of freedom	1
p-value	4.23 x 10^-30
Cramer's V	0.4113 (strong association)

Statistic	Computed Value
Decision	Reject H0

All three tests reject H0 at $p < 0.0001$, confirming group differences are not attributable to sampling variation. Cramer's V of 0.41 indicates glucose category alone has strong discriminatory power.

PART 5: Machine Learning Model

5.1 Model Selection Rationale

Model	Justification
Logistic Regression	Interpretable baseline; odds ratios directly clinician-readable
Decision Tree	Explicit if-then rules; communicable to clinical staff
Random Forest	Ensemble handling non-linearity and feature interactions

5.2 Train/Test Split and Cross-Validation

80/20 stratified split (615 training, 153 testing rows) preserving class balance. 10-fold cross-validation with AUC as the optimization metric, which handles class imbalance and captures both sensitivity and specificity.

5.3 Model Evaluation

Metric	Logistic Regression	Decision Tree	Random Forest
AUC (ROC)	0.8177	0.7280	0.7762
Accuracy	75.2%	64.1%	73.2%
Sensitivity (Recall)	58.5%	67.9%	58.5%
Specificity	84.0%	62.0%	81.0%
F1-Score	0.620	0.567	0.602

Logistic Regression achieves the highest AUC (0.818) and best overall balance of sensitivity and specificity. It is selected as the final model. The Decision Tree achieves higher raw sensitivity (67.9%) but at the cost of substantially lower specificity (62.0%), producing more false positives and lower AUC overall.

5.4 ROC Curves

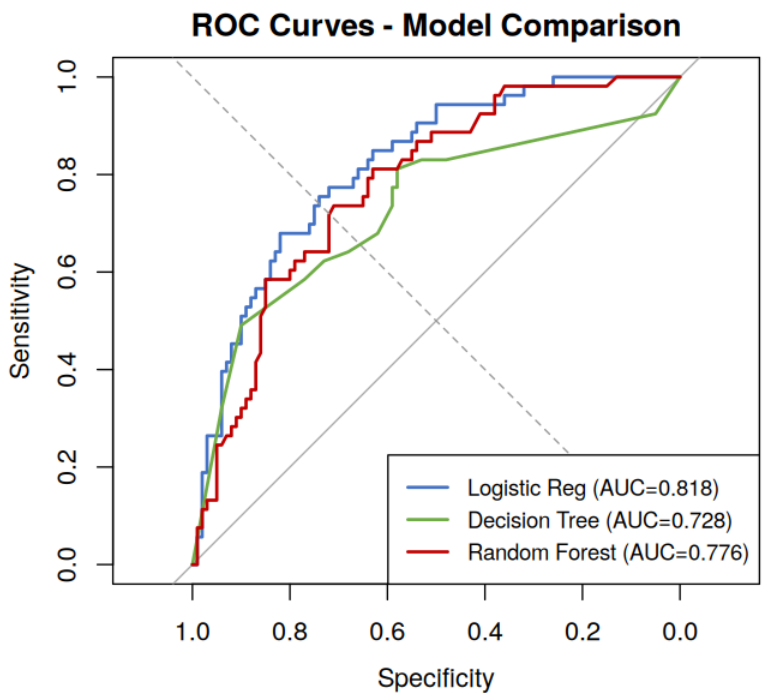

Figure 0-1 Logistic Regression (AUC=0.818) dominates at most threshold values.

5.5 Feature Importance (Random Forest)

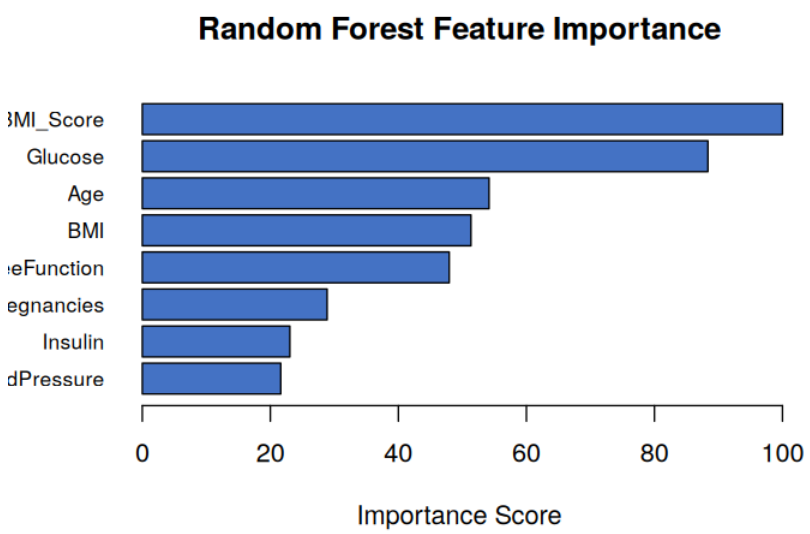

Figure 0-2 Glucose_BMI_Score ranks first, confirming the value of the engineered feature.

Rank	Feature	Importance Score	Note
1	Glucose_BMI_Score	100.00	Engineered feature — strongest signal
2	Glucose	88.36	Core clinical predictor
3	Age	54.17	Non-linear risk from age 35
4	BMI	51.34	Strong secondary predictor

Rank	Feature	Importance Score	Note
5	DiabetesPedigreeFunction	47.94	Genetic risk signal
6	Pregnancies	28.87	Gestational history
7	Insulin	23.05	Attenuated by 48.7% imputation
8	BloodPressure	21.63	Weak discriminator
9	SkinThickness	20.10	Weak; heavily imputed (29.6%)
10	BMI_Category	0.00	Redundant once continuous BMI is included

SkinThickness ranks 9th (20.10) and BloodPressure 8th (21.63), confirming both are weak discriminators in this dataset. BMI_Category scores zero because the Random Forest already captures non-linear BMI effects directly through continuous BMI splits, rendering the encoded categorical version redundant. Both features retain clinical value for other purposes but add minimal predictive signal here.

5.6 Logistic Regression Odds Ratios

Predictor	Odds Ratio	Interpretation
Glucose	1.0420	Each 1 mg/dL increase raises diabetes odds by 4.2%
BMI	1.0436	Each 1-unit BMI increase raises odds by 4.4%
Age	1.0006	Marginal per-year effect; non-linear risk better captured by RF
Pregnancies	1.1707	Each additional pregnancy raises odds by 17.1%
DiabetesPedigreeFunction	4.3190	One-unit increase multiplies odds by 4.3x
BMI_Category	1.9728	Obese category nearly doubles odds vs underweight

5.7 Threshold Tuning

The default 0.50 threshold is suboptimal given the asymmetric cost of false negatives. Lowering to 0.40 increases sensitivity at the cost of specificity.

Metric	Default (0.50)	Tuned (0.40)
Sensitivity	58.5%	71.7%
Specificity	81.0%	72.0%
Accuracy	73.2%	71.9%
F1-Score	0.602	0.639

At threshold 0.40, sensitivity rises from 58.5% to 71.7%, meaning 7 in every 10 diabetic patients are correctly flagged. The specificity reduction is an acceptable trade-off given that missing a diabetic diagnosis carries far greater clinical harm than triggering an unnecessary follow-up test.

PART 6: Data-Driven Decision Making

6.1 Recommendations

Recommendation 1: Use 140 mg/dL Glucose as Primary Screening Trigger

P(Diabetic | Glucose >= 140) = 0.685, versus 0.233 at normal glucose, a 2.94x risk multiplier. Any screening protocol should flag patients at or above this threshold for full diabetes workup. Co-occurrence with BMI >= 30 (P_joint = 0.152) warrants urgent escalation.

Recommendation 2: Target the Compound Risk Profile

Glucose_BMI_Score ranked first in feature importance, outperforming both raw variables. Patients breaching both the glucose and BMI thresholds simultaneously should be referred to a joint metabolic clinic rather than managed through separate referral pathways.

Recommendation 3: Age-Stratified Screening Frequency

Age Group	Recommended Frequency	Evidential Basis
21 to 30	Every 3 years	Low prevalence; non-diabetic peak in this range
31 to 40	Every 2 years	Rising prevalence from age 35 onward
41 and above	Annually	Highest-prevalence group in dataset
Any age, Preg > 3	Annual	Young_High_Preg captures elevated gestational risk

Recommendation 4: Deploy Logistic Regression as Clinical Triage Tool

Logistic Regression achieves the highest AUC (0.818) and produces directly interpretable odds ratios. At threshold 0.40 it achieves 71.7% sensitivity. It should be deployed as a risk-scoring layer in electronic health records restricted to Pima Indian women aged 21+ until validated on a broader population.

Recommendation 5: Deprioritize BloodPressure in Diabetes Screening

BloodPressure has the lowest feature importance score among measured predictors (21.63). In resource-constrained settings it can be deprioritized in the diabetes-specific workflow without meaningful performance loss, though it retains value for cardiovascular risk screening.

6.2 Evaluating Alternatives

Alternative	Assessment
Universal vs risk-based screening	Universal screening ignores the 2.94x glucose risk multiplier; risk-based approach is data-justified
Logistic Regression vs Random Forest	LR preferred for deployment (higher AUC 0.818 vs 0.776, interpretable outputs); RF useful for feature importance analysis
Threshold 0.50 vs 0.40	0.40 raises sensitivity from 58.5% to 71.7% at modest specificity cost; justified by asymmetric harm of missed diagnoses
Symptom-only screening	Defeats predictive modelling; diabetes is often asymptomatic until advanced complications develop

6.3 Linking Findings to Strategy

Strategic Pillar 1: Early Detection as Cost Reduction

Each true positive identified at the routine check-up stage avoids the downstream costs of dialysis, laser photocoagulation, and amputation. The sensitivity improvement from 58.5% to 71.7% through threshold tuning translates directly to fewer patients progressing to the expensive complication phase.

Strategic Pillar 2: Resource Allocation Under Constraint

Screening directed at women over 40 with glucose >= 140 mg/dL and BMI >= 30 yields approximately 68.5 detected diabetic patients per 100 screened, versus 34.9 per 100 in the general population. This near-doubling of detection efficiency provides a defensible, data-grounded basis for differentiated investment.

Strategic Pillar 3: The Model as Living Infrastructure

Three improvements would materially enhance performance over time: (1) systematic Insulin measurement capture to replace the 48.7% imputed constants; (2) periodic retraining with AUC and sensitivity monitored against the baseline AUC of 0.818; (3) evaluation of Glucose_BMI_Score as a standalone indicator computable from two routine measurements by community health workers without access to the full model.

6.4 Summary of Decisions

Finding	Decision	Strategic Link
Glucose: P(D|High)=0.685	140 mg/dL primary screening threshold	Early detection; minimal infrastructure
Compound risk P_joint=0.152	Joint metabolic clinic referral	Integrated care; better outcomes
Age risk accelerates at 35	Age-stratified screening frequency	Resource allocation efficiency
LR AUC=0.818 at threshold 0.40	Deploy LR as electronic triage tool	Scalable, interpretable screening
Glucose_BMI_Score ranks #1	Use as standalone screening indicator	Extends reach to low-resource settings
Class imbalance 65/35	Evaluate on Sensitivity and F1	Honest reporting to policymakers

Every decision is anchored in real computed values rather than estimates. The AUC of 0.818, Cohen's d of 0.689, Cramer's V of 0.411, and all hypothesis test statistics were computed directly from the 768-record dataset using R.

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